MILITARY SPECIFICATION
RECEIVER, RADIO R-1651/ARA

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be of one type and shall provide for the reception of signals from radio sonobuoys to provide sonobuoy homing and on-top position indicating when used in conjunction with direction finder equipment.

2. APPLICABLE DOCUMENTS

2.1 General- The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-T-5422	Environmental Testing Aircraft Electronic Equipment
MIL-D-7033	Direction Finder Group AN/ARA-25, Military Specification

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures: Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-23103	Thermal Performance Evaluation Airborne Electronics Equipment, General Requirements for

Naval Air Systems Command

AR-34	Failure Classification for Reliability Testing, General Requirements for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipments
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-781	Reliability Test, Exponential Distribution

2.1.1 Availability of Documents - (1) When requesting specifications, Standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to specification MIL-E-5400.

(2) Non-repairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Non-repairable subassemblies must be reliable (See 6.5).

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, exluding cables and mount, shall be a minimum consistent with good design and shall not exceed 3.5 pounds.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

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3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure 9MTBF) - The equipment shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements 4.4.3.

3.3.3 Cabling and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet the interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

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3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-STD-461.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Test points shall be provided on the front of the receiver to monitor significant voltages necessary to the maintenance cycle, including as a minimum:

a. Direction Finder Group Power Supply
b. Receiver Power Supply
c. Audio Output
d. AGC
e. Ground

Additional internal test points as necessary shall be provided for ease of trouble analysis.

3.3.7 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, namplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC, 400 Hz and 27.5 ± 0.5 VDC

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3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of Specification MIL-T-5422.

3.3.9.2 Altitude - The receiver shall be capable of satisfactory operation at altitudes up to 35,000 feet.

3.3.10 Warm-up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed 5 seconds under standard conditions and 15 seconds at extreme service conditions.

3.3.11 Input Electrical Power -

3.3.11.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits defined in MIL-STD-704.

(1) AC Power (Single Phase), 115 V, Category B, 15VA
(2) DC Power, 28V, Category B, 20 Watts

3.3.12 Output Power - The receiver shall be capable of continuously providing 225 volts DC $\pm$ 10 percent at 30 milliamperes $\pm$ 20 percent tube plate output power for either the AN-608/ARA-25 or AM-1777/ARA-25A Electronic Control Amplifiers, except that the equipment shall provide 225 volts DC $\pm$ 40 percent during the normal voltage transients of MIL-STD-704. With normal steady state voltage applied, this voltage shall contain no more than 0.50 volt ripple (RMS).

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3.3.13 Internal Adjustments - Adjustments shall be kept to a minimum and shall be easily accessible upon removal of receiver dust cover. All adjustments shall be factory aligned and present prior to delivery so that the receiver is operating satisfactorily to the requirements herein. Alignment shall be stable for a minimum of 400 normal operation hours.

3.3.14 Dust Cover - An external dust cover shall be supplied which is readily removable without attachment to wiring or other obstructions. The dust cover shall provide adequate sand and dust protection as specified by MIL-E-5400.

3.3.15 Mount - A mounting plate providing for easy attachment and removal of the receiver shall be supplied.

3.3.16 Cooling - The equipment shall operate satisfactorily under the thermal conditions determined from an evaluation per MIL-T-23103.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The receiver shall be compatible with, match impedance with, and accept the voltage and impedance output levels from the AS-578/ARA-25 or AS-578A/ARA-25 antenna assemblies. Antenna sharing capability shall be provided by an internal miniature coaxial relay. Relaying of tube plate power for the direction finder group shall be provided internally. Associated systems functions including AGC, sensitivity, audio and ADF shall also be switched as necessary. All receiver connectors as well as all circuit protective devices, if required, shall be located on the front of the unit. The receiver shall preamplify, convert, I.F. amplify, and amplitude detect the antenna output and provide acceptable voltage and impedance output

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levels to be compatible with the input requirements of the Electronic Control Amplifiers AM-1777/ARA-25A or AM-608/ARA-25. The receiver shall operate in conjunction with the AN/ARA-25 Direction Finder Group to provide a sonobuoy homing capability as displayed on an appropriate bearing, heading indicator. The AN/ARA-25 power supply shall be contained as a separate module, and shall be removable without affecting the overall receiver function. (See MIL-D-7033).

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3.4.2 Frequency Range - The receiver shall operate over the range of frequencies from 162 to 174 MHz in such manner that any one of the following thirty-one channels may be switch-selected.

Channel No.	Frequency (MHz)	Channel No.	Frequency (MHz)
1	162.250	17	162.625
2	163.000	18	163.375
3	163.750	19	164.125
4	164.500	20	164.875
5	165.250	21	165.625
6	166.000	22	166.375
7	166.750	23	167.125
8	167.500	24	167.875
9	168.250	25	168.625
10	169.000	26	169.375
11	169.750	27	170.125
12	170.500	28	170.875
13	171.250	29	171.625
14	172.000	30	172.375
15	172.750	31	173.125
16	173.500		

3.4.3 Local Oscillator - The local oscillator stability shall be such that any change in alignment, including reset error caused by cycling of the channel selector or drift caused by aging of component parts, shall not exceed ± 10 kilohertz from the assigned channel frequency for the temperature range from -20°C to °35°C. For the temperature ranges from -54°C to -20°C and from +35°C to +55°C, the local oscillator drift shall not exceed ± 15 kilohertz from the assigned channel frequency.

3.4.4 Overall Gain Stability - The gain stability of this receiver shall be such that regenerative oscillatory conditions do not prevail for any value of carrier level at the antenna connector below 10,000 microvolts.

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3.4.5 Sensitivity - A 10-decibel (db) signal-to-noise ratio shall exist at the audio output when a 2.5-microvolt signal, 30 percent modulated by a 100 Hz square wave, is applied at the antenna connector.

3.4.6 Noise Figure - The noise figure of this equipment shall be no greater than 12 db. A nominal value shall be 6 db.

3.4.7 Single-Channel Bandwidth - For any given channel, the receiver minimum 6-db bandwidth shall be no less than 190 kHz and no more than 255 kHz. The slope factor shall provide a minimum of 60-db adjacent channel rejection.

3.4.8 IF Rejection - The receiver shall attenuate the IF center frequency at least 80 db relative to any desired channel center frequency applied at the receiver input.

3.4.9 Image and Other Spurious Responses - The receiver shall attenuate image and other spurious responses external to the passband at least 60 db compared to any desired channel center frequency except at one-half signal frequency the response shall be down at least 45 db. There shall be no more than two spurious responses in any channel within the passband which are less than 60 db down, and these shall be at least 50 db down.

3.4.10 RF Input Protection - The receiver shall be suitably protected for input signals up to 500,000 microvolts at any frequency or combination of frequencies below 30,000 MHz.

3.4.11 Audio Output Characteristics -

3.4.11.1 Frequency Response - The frequency response of the receiver shall be such that the audio output from 75-500 Hz is flat within $\pm$ 2 db of the reference level output at 100 Hz when measured over an external 22,000-ohm resistive impedance.

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3.4.11.2 Audio Level - The root-mean-square (rms) level of audio output for 100 Hz shall be 0.5 volt .

3.4.11.3 Audio Level Range - When a 100 Hz, 30-percent square wave modulated carrier signal of 2.5 microvolts or greater is applied to the receiver, the amplitude detected and amplified audio output shall be adjustable from 0.25 volt rms minimum to 1.0 volt rms over an external 22,000-ohm resistive impedance.

3.4.12 Overall Phase Delay - Phase delay between the 100-Hz modulation entering the receiver and the 100-Hz demodulated audio signal used as the input to the Electronic Control Amplifiers of the AN/ARA-25 Direction Finder Group shall be 180 ± 10 degrees.

3.4.13 RF Bandpass Filter - RF amplifier circuits shall feature an input and output tuned circuit or filter to reject signals below 162.25 megahertz and above 173.50 megahertz.

3.4.14 Reduced Size Coaxial Relay - Internal coaxial relay used for antenna sharing shall have a maximum of 0.5-db insertion loss and shall have a minimum of 40-db isolation between open contacts measured at 225 megahertz.

3.4.15 Remote "Adequate Signal Strength Indicator" - The receiver shall contain switching circuits and a threshold control that is adjustable and shall operate upon receiving an RF signal of -86 dbm at the receiver antenna connector. It shall never operate in the absence of signal. When actuated, the switching circuits shall supply a ground to the external circuit. Depending upon circuit design, performance of the above function without the use of a threshold control is preferred.

3.4.16 Headphone Output - A nominal audio output of 4 to 6 volts (rms) shall be supplied at 400 Hertz. The output audio line shall be 600 ohms.

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3.4.17 Radio Interference Suppression - Radio interference suppression shall conform to the requirements of MIL-I-6181. The receiver shall include radio frequency (RF) bandpass features in the RF amplifier and adequate individual channel selectivity in succeeding stages.

3.4.18 Form Factor - The equipment shall be enclosed in a single case not to exceed 3.05 inches wide by 4.95 inches high by 5.30 inches long. The receiver shall be hard mounted without vibration isolators.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory

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acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as

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a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in test.

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following.

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1#
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

One out of first ten need not be selected and tested if initial production and test are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possiblity of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

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4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept/Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test Level E shall be used.

4.4.3.2.1 All Equipment Test - Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for 70 hours. Prior to the 70-hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan XXIX of MIL-STD-781. (Extend the line as necessary to accomodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan XXIX curves show a rejection situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

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4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. The procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5. Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room Ambient
AC Voltage	115 ± 5 volts (at lowest applicable frequency)
DC Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

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4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting pre-production tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Test Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking,the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning the original shall be furnished the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555 the method of preservation for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - This equipment will be used in Navy ASW aircraft to provide sonobuoy homing and on-top position indicating. It will serve the same purpose as Receiver, Radio R-1047A/A with 31 channels of radio frequency. These frequencies are spaced 375 kHz apart from 162.250 MHz to 173.500 MHz.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (Such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

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6.7 This specification is under the cognizance of AIR-533014.

Project Number 5845-N044

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-81680(AS) RECEIVER, RADIO R-1651/ARA

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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